MTH 201 Infinite Series (focused on Alternating Series)

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Determine if the series is absolutely convergent, conditionally convergent, or divergent.

a) $\sum_{n=1}^{\infty}(-1)^n \frac{2}{\sqrt[3]{n}}$

Note $\sum_{n=1}^{\infty}\left|(-1)^n \frac{2}{\sqrt[3]{n}}\right| = \sum_{n=1}^{\infty}\frac{2}{\sqrt[3]{n}} = \sum_{n=1}^{\infty}\frac{2}{n^{1/3}}$ and this is a p-series with $p=\frac{1}{3}<1$ and therefore diverges. This means that the given series is absolutely divergent.

However, the given series is an alternating series with $a_n = \frac{2}{\sqrt[3]{n}}$. Clearly a_n is decreasing and $\lim_{n\to\infty} a_n = 0$ which implies, by the AST, that the given series converges.

This means that the series $\sum_{n=1}^{\infty}(-1)^n \frac{2}{\sqrt[3]{n}}$ is conditionally convergent.

b) $\sum_{n=2}^{\infty}(-1)^n ne^{-n}$

Note $\sum_{n=2}^{\infty}\left|(-1)^n ne^{-n}\right| = \sum_{n=2}^{\infty}\frac{n}{e^n}$ and $\frac{n}{e^n} < \frac{2^n}{e^n} = \left(\frac{2}{e}\right)^n$. The series $\sum_{n=2}^{\infty}\frac{n}{e^n}$ and $\sum_{n=2}^{\infty}\left(\frac{2}{e}\right)^n$ are series with non-negative terms and $\sum_{n=2}^{\infty}\left(\frac{2}{e}\right)^n$ is a convergent geometric series with $r=\frac{2}{e}$. Therefore, by the Comparison Test, the given series is absolutely convergent.

c) $\displaystyle\sum_{n=1}^{\infty} \frac{\sqrt{n \arctan(n)}}{n^{1/2}}$

Note that $\displaystyle\sum_{n=1}^{\infty} \frac{\sqrt{n \arctan(n)}}{n^{1/2}} = \sum_{n=1}^{\infty} \frac{\sqrt{n \arctan(n)}}{\sqrt{n}} = \sum_{n=1}^{\infty} \sqrt{\frac{n \arctan(n)}{n}} = \sum_{n=1}^{\infty} \sqrt{\arctan(n)}$.

Also, $\displaystyle\lim_{n\to\infty} \sqrt{\arctan(n)} = \sqrt{\frac{\pi}{2}}$.

Therefore, by the Test for Divergence, the given series diverges.

d) $\displaystyle\sum_{n=0}^{\infty} \frac{n^2 - n + 1}{\sqrt{n^5 + 300n^3 + 2}}$

We would like to do a limit comparison with the series $\displaystyle\sum_{n=0}^{\infty} \frac{1}{n^{1/2}}$ but notice that this series is not defined when $n = 0$. Therefore, we note observe the series that $\displaystyle\sum_{n=1}^{\infty} \frac{n^2 - n + 1}{\sqrt{n^5 + 300n^3 + 2}}$ and $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$. Both series contain only positive terms and we calculate that

$$\lim_{n\to\infty} \frac{\left(\dfrac{n^2 - n + 1}{\sqrt{n^5 + 300n^3 + 2}}\right)}{\left(\dfrac{1}{n^{1/2}}\right)} = \lim_{n\to\infty} \frac{n^{5/2} - n^{3/2} + n^{1/2}}{\sqrt{n^5 + 300n^3 + 2}} = \lim_{n\to\infty} \frac{n^{5/2} - n^{3/2} + n^{1/2}}{\sqrt{n^5 + 300n^3 + 2}} \cdot \frac{1/n^{5/2}}{1/n^{5/2}}$$

$$= \lim_{n\to\infty} \frac{1 - \dfrac{1}{n} + \dfrac{1}{n^2}}{\dfrac{\sqrt{n^5 + 300n^3 + 2}}{\sqrt{n^5}}} = \lim_{n\to\infty} \frac{1 - \dfrac{1}{n} + \dfrac{1}{n^2}}{\sqrt{1 + \dfrac{300}{n^2} + \dfrac{2}{n^5}}} = 1$$

Since the result is a positive, finite number then by the Limit Comparison Test the series $\displaystyle\sum_{n=1}^{\infty} \frac{n^2 - n + 1}{\sqrt{n^5 + 300n^3 + 2}}$ and $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$ must have the same behavior. Note that $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$ is a p-series with $p = \dfrac{1}{2} < 1$ and therefore diverges. Then since $\displaystyle\sum_{n=1}^{\infty} \frac{n^2 - n + 1}{\sqrt{n^5 + 300n^3 + 2}}$ and $\displaystyle\sum_{n=0}^{\infty} \frac{n^2 - n + 1}{\sqrt{n^5 + 300n^3 + 2}}$ differ by only a single term it must be the case that $\displaystyle\sum_{n=0}^{\infty} \frac{n^2 - n + 1}{\sqrt{n^5 + 300n^3 + 2}}$ also diverges.

2. Is it true that if $\sum_{n=1}^{\infty} a_n$ is divergent then $\sum_{n=1}^{\infty} |a_n|$ is also divergent?

Yes! If $\sum_{n=1}^{\infty} |a_n|$ was convergent then $\sum_{n=1}^{\infty} a_n$ would also have to be convergent.

3. What if $\sum_{n=1}^{\infty} |a_n|$ is convergent; is $\sum_{n=1}^{\infty} a_n$ also convergent?

Yes! Any absolutely convergent series must be convergent.

4. If $\sum_{n=1}^{\infty} a_n$ is convergent is $\sum_{n=1}^{\infty} |a_n|$ also convergent?

No! Consider, for example, the alternating harmonic series which is convergent, $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ but absolutely divergent.

5. Consider the series $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n^2}$

a) Show that the given series is convergent.

Note that this is an alternating series with $a_n = \frac{1}{n^2}$. Clearly a_n is decreasing and $\lim_{n\to\infty} a_n = 0$. Therefore, by the AST, the series converges.

b) While we know the given series is convergent we don't know its final value. Find s_{10} to use it as an estimate of what the series converges to.

Suppose that $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n^2} = S$, then

$$S \approx \sum_{n=1}^{10} (-1)^n \frac{1}{n^2} = \frac{-1}{1} + \frac{1}{4} - \frac{1}{9} + \frac{1}{16} - \frac{1}{25} + \frac{1}{36} - \frac{1}{49} + \frac{1}{64} - \frac{1}{81} + \frac{1}{100} = -\frac{5194287}{6350400} \approx -0.81796$$

c) Clearly the series doesn't converge to the value of s_{10}, but it should converge to something close to this value. Find an upper bound on the error of the approximation of s_{10} as the sum of the series.

Since the given series is a convergent alternating series we know that
$|S - s_{10}| \le a_{11} = \frac{1}{(11)^2} = \frac{1}{121}$.

This means that the value we found in part (b) is, at most, $\frac{1}{121} \approx 0.008$ units away from the correct sum of the series.

d) Suppose that we wanted to estimate the sum of the series accurately to within $\frac{1}{1{,}000{,}000}$ of its actual value. How many terms of the series would we need to use in a partial sum of the series to do this?

We want to determine a value of n such that $|S - s_n| \le a_{n+1} \le \frac{1}{1{,}000{,}000}$.

We note that $a_{1000} = \frac{1}{(1000)^2} = \frac{1}{1{,}000{,}000}$. This means that if we let $n+1 = 1000 \;\rightarrow\; n = 999$ then we will have an approximation of S that is, at most off by $\frac{1}{1{,}000{,}000}$.

So we need to calculate s_{999}, that is the 999$^{\text{th}}$ partial sum of the series.

e) It can be shown that the exact sum of this series is $-\frac{\pi^2}{12}$. Use Wolfram Alpha to show that the differences between your answers to parts (c) and (d) and the actual sum of the series are indeed accurate to the appropriate level.

We are being told that $S = -\frac{\pi^2}{12}$.

In part (c) we estimated this value with $s_{10} = -\frac{5194287}{6350400}$. The difference between these values is roughly 0.0045. This means that our estimation of s_{10} was definitely within $\frac{1}{121} \approx 0.008$ units away from the correct value of the series.

In part (d) we estimated the value of S with s_{999}. The difference between these values is roughly 0.0000005005. This means that our estimation of s_{999} was definitely within $\frac{1}{1{,}000{,}000}$ units away from the correct value of the series.